

Rapid-Charging MICRO-SUPERCAPACITORS For Portable Devices



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Batteries are the dominant technology for powering portable devices. But the problem with batteries is their size and requirement of frequent charging, time required for charging, and repair and replacement.

Fig. 1 shows the charging time for the batteries used in some mobile phones. These batteries discharge within hours of use as shown in Fig. 2. Capacitors are also energy storage elements that, unlike batteries, generate an electrical field between two parallel conductor plates. The energy stored in a capacitor depends on the value of its capacitance and the voltage applied across its plates. For the same voltage the capacitor will store more energy for a large value of capacitance. Since the capacitance of a capacitor is directly proportional to surface area of its electrodes and inversely proportional to the distance between electrodes, it requires a large area to store charge.

The current trend in portable devices is continuous miniaturisation, while ensuring the functionality and reliability of existing components. The main problem faced in miniaturisation is the

integration of batteries as energy storage elements. The size of energy storage elements often limits the miniaturisation of the entire system. This is because the necessary energy-storage components cannot be scaled down to the size of electronic gadgets. So designing efficient miniaturised energy storage devices for energy delivery or harvesting with high-power capabilities remains a challenge.

Supercapacitors or ultracapacitors

An improvement in energy storage devices has been the development of electric double-layer capacitor (EDLC), best known as 'supercapacitor' or 'ultracapacitor'. A supercapacitor has large-surface-area conductive plates (electrodes), placed very close to each other.

Supercapacitors provide many advantages over batteries. For instance, these have a long cycle lifetime—they can be cycled hundreds of thousands times without much change in performance. The lifespan of a supercapacitor is 10 to 20 years, with a possible capacity reduction from the original 100 per cent to 80 per cent after ten years of usage or so. Supercapacitors have a low equivalent series resistance. These can therefore provide a high power density and high load currents to achieve almost instant charge in seconds.

The new hybrid supercapacitors store large amounts of energy, recharge quickly and last more than 10,000 recharge cycles. This is a great advantage for smartphones, which will charge in just 30 seconds and then continue working through the rest of the day on a single charge.

The basic principle of energy storage in supercapacitors is the same as that of ordinary capacitors. However,

Fig. 1: Charging time of different batteries used in mobile phones

